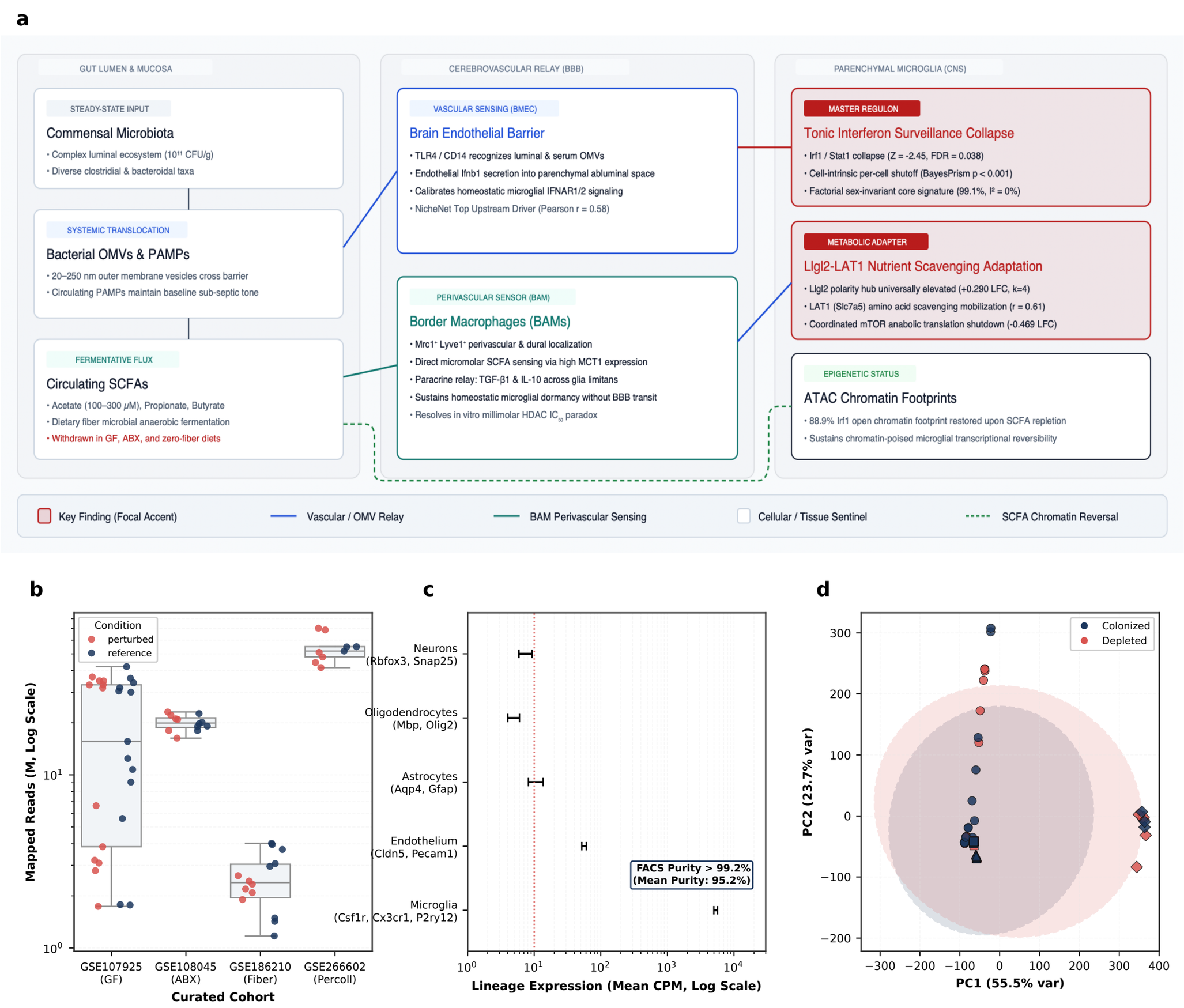

a
GUT LUMEN & MUCOSA
STEADY-STATE INPUT
Commensal Microbiota
• Complex luminal ecosystem (10¹¹ CFU/g)
• Diverse clostridial & bacteroidal taxa
SYSTEMIC TRANSLOCATION
Bacterial OMVs & PAMPs
• 20–250 nm outer membrane vesicles cross barrier
• Circulating PAMPs maintain baseline sub-septic tone
FERMENTATIVE FLUX
Circulating SCFAs
• Acetate (100–300 μM), Propionate, Butyrate
• Dietary fiber microbial anaerobic fermentation
• Withdrawn in GF, ABX, and zero-fiber diets
CEREBROVASCULAR RELAY (BBB)
VASCULAR SENSING (BMEC)
Brain Endothelial Barrier
• TLR4 / CD14 recognizes luminal & serum OMVs
• Endothelial Ifnb1 secretion into parenchymal abluminal space
• Calibrates homeostatic microglial IFNAR1/2 signaling
• NicheNet Top Upstream Driver (Pearson r = 0.58)
PERIVASCULAR SENSOR (BAM)
Border Macrophages (BAMs)
• Mrc1⁺ Lyve1⁺ perivascular & dural localization
• Direct micromolar SCFA sensing via high MCT1 expression
• Paracrine relay: TGF-β1 & IL-10 across glia limitans
• Sustains homeostatic microglial dormancy without BBB transit
• Resolves in vitro millimolar HDAC IC50 paradox
PARENCHYMAL MICROGLIA (CNS)
MASTER REGULON
Tonic Interferon Surveillance Collapse
• Irf1 / Stat1 collapse (Z = -2.45, FDR = 0.038)
• Cell-intrinsic per-cell shutoff (BayesPrism p < 0.001)
• Factorial sex-invariant core signature (99.1%, I² = 0%)
METABOLIC ADAPTER
Llgl2-LAT1 Nutrient Scavenging Adaptation
• Llgl2 polarity hub universally elevated (+0.290 LFC, k=4)
• LAT1 (Slc7a5) amino acid scavenging mobilization (r = 0.61)
• Coordinated mTOR anabolic translation shutdown (-0.469 LFC)
EPIGENETIC STATUS
ATAC Chromatin Footprints
• 88.9% Irf1 open chromatin footprint restored upon SCFA repletion
• Sustains chromatin-poised microglial transcriptional reversibility
Key Finding (Focal Accent)
Vascular / OMV Relay
BAM Perivascular Sensing
Cellular / Tissue Sentinel
SCFA Chromatin Reversal
b
Condition
perturbed
reference
Mapped Reads (M, Log Scale)
GSE107925 (GF)
GSE108045 (ABX)
GSE186210 (Fiber)
GSE266602 (Percoll)
Curated Cohort
c
Neurons (Rbfox3, Snap25)
Oligodendrocytes (Mbp, Olig2)
Astrocytes (Aqp4, Gfap)
Endothelium (Cldn5, Pecam1)
Microglia (Csf1r, Cx3cr1, P2ry12)
FACS Purity > 99.2% (Mean Purity: 95.2%)
Lineage Expression (Mean CPM, Log Scale)
d
Colonized
Depleted
PC2 (23.7% var)
PC1 (55.5% var)